Contrast-Enhanced Emotion Recognition Using Lightweight CNNs

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Motivation & Problem Statement

Importance of Emotions

Emotions are key for human-machine interaction enhancements.

Challenges

- Subtle emotional variations
- Low image resolution
- Noisy datasets

FER2013 Dataset

Consists of 48x48 grayscale images with label noise challenges.

Motivation & Problem Statement

1. **Emotions are essential for human interaction**

   Automatically recognizing emotions can greatly enhance user experiences in a wide range of applications, from human-computer interaction to customer service.

2. **Facial Emotion Recognition (FER)**

   FER allows machines to interpret emotions from visual cues in facial expressions, providing a natural and intuitive way to understand human behavior.

3. **Major Challenges**

   Key challenges in FER include subtle emotional variations, low image resolution, and noisy or imbalanced datasets.

Objectives

Model Implementation

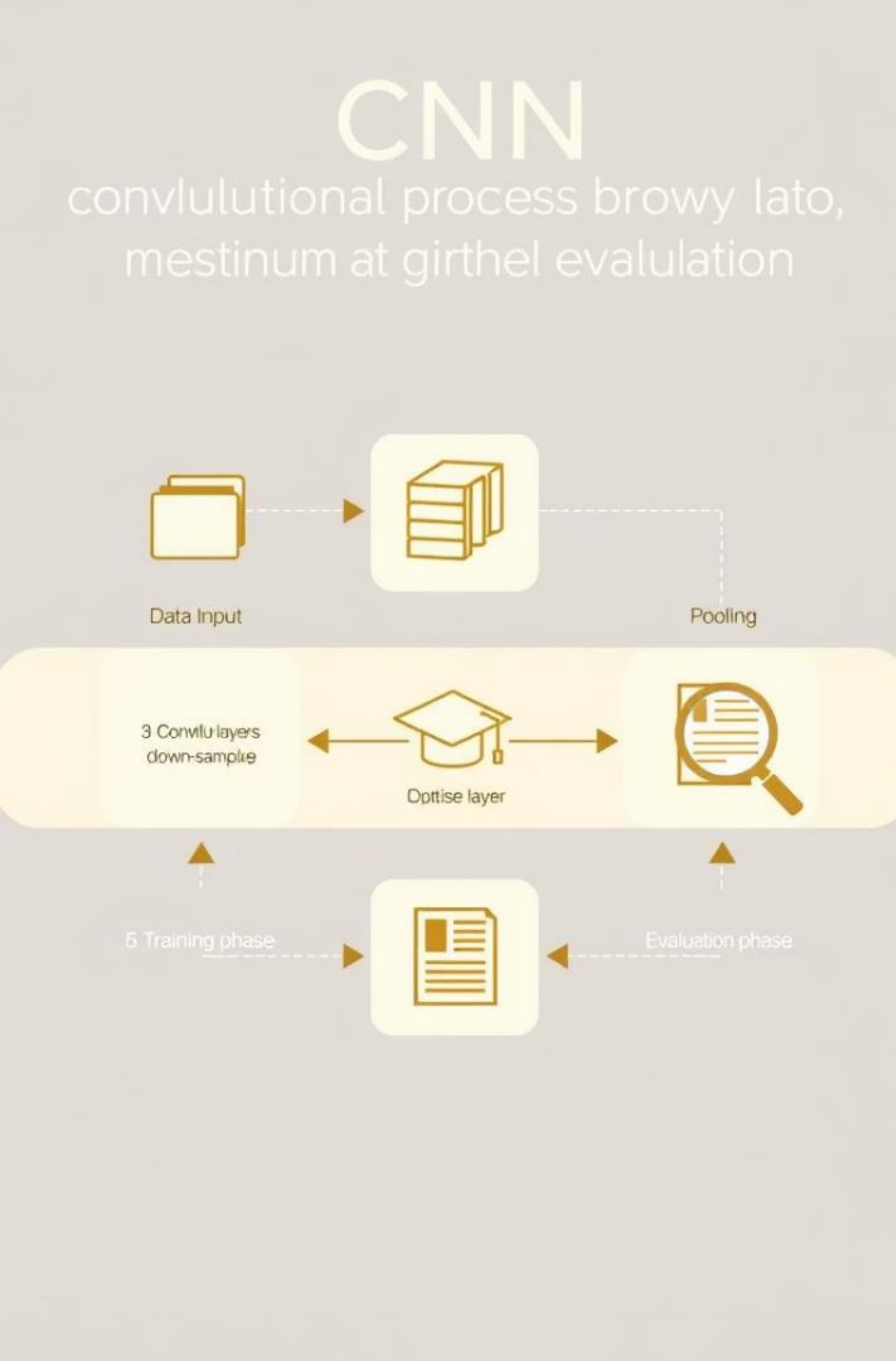
Custom lightweight CNN and pretrained ResNet18.

Image Enhancement

CLAHE to improve local contrast in images.

Performance Analysis

Evaluate accuracy, stability, and class-wise metrics.



Related Work

VGGNet Transfer Learning

Khaireddin & Chen (2021) achieved 73.28% accuracy.

Handcrafted + Deep Features

Georgescu et al. (2018) improved to 75.42% accuracy.

CNN Ensembles

Stanford CS230 (2020) used ensemble methods for boost.

Lightweight CNNs

Gursesli et al. (2024) focused on edge device optimization.

Hybrid Systems

Oguine et al. (2022) combined deep learning with Haar classifiers.

Dataset & Preprocessing

Dataset Details

- 35,887 images sized 48x48 pixels
- 7 emotion classes including angry, happy, sad
- Grayscale format impacts feature extraction

Preprocessing Steps

- Image resizing and normalization
- CLAHE for local contrast enhancement
- Data augmentation: flips and rotations

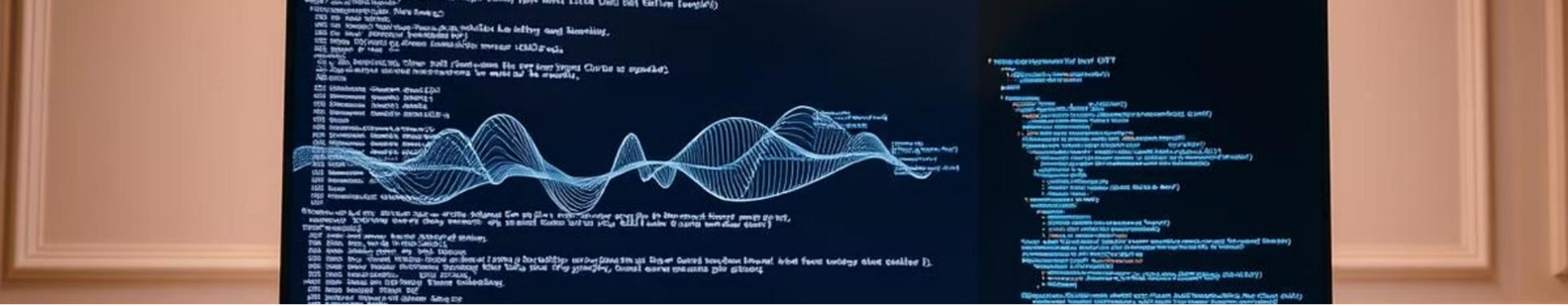
Model Architectures

Custom Lightweight CNN

- 3 convolutional layers with BatchNorm and ReLU
- MaxPooling, dropout to reduce overfitting
- Fully connected output layer for 7 emotions

Pretrained ResNet18

- Trained on ImageNet, modified for grayscale input
- Fine-tuning of final and deeper layers
- 7-class output layer adjustment



Training Setup

Optimizer & Loss

Adam optimizer with CrossEntropy loss and label smoothing.

Training Details

Trained for 25 epochs, balancing speed and convergence.

Evaluation Metrics

Test accuracy, confusion matrix, and top-3 prediction accuracy.

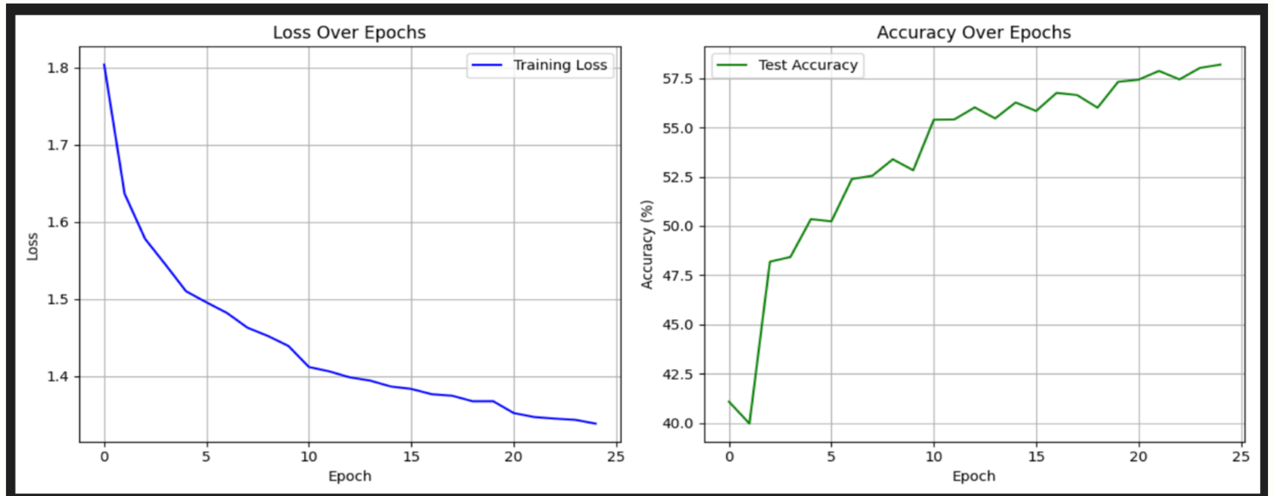
Results - Baseline CNN

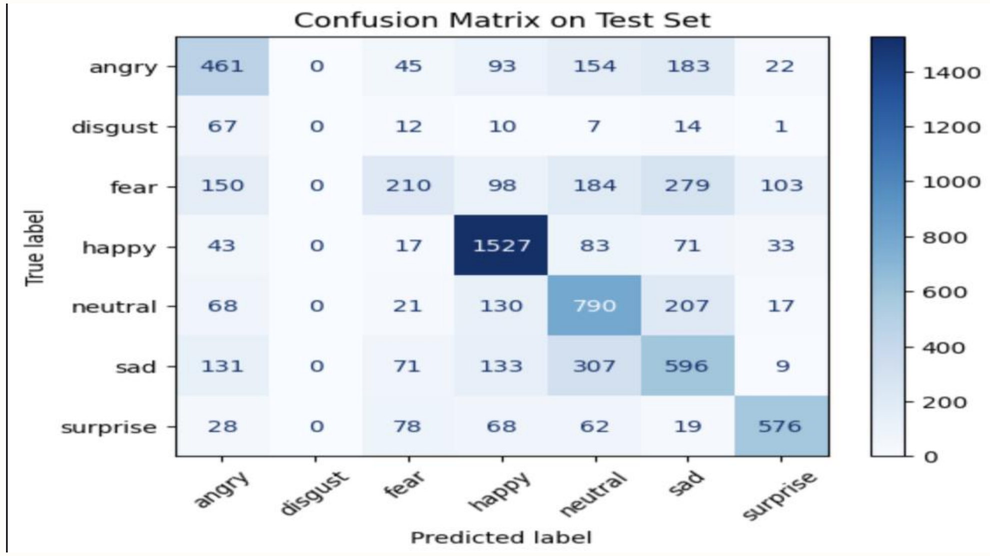
Performance

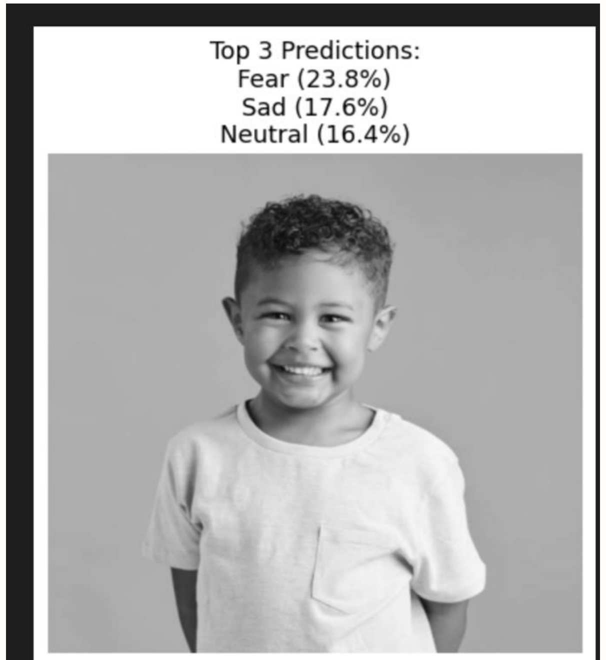
- Test accuracy: 58.58%
- Strong on 'happy' emotion
- Weak on 'fear' and 'disgust'

Training Behavior

Consistent accuracy and loss improvements across epochs.







Results - Custom CNN + CLAHE

Accuracy & Stability

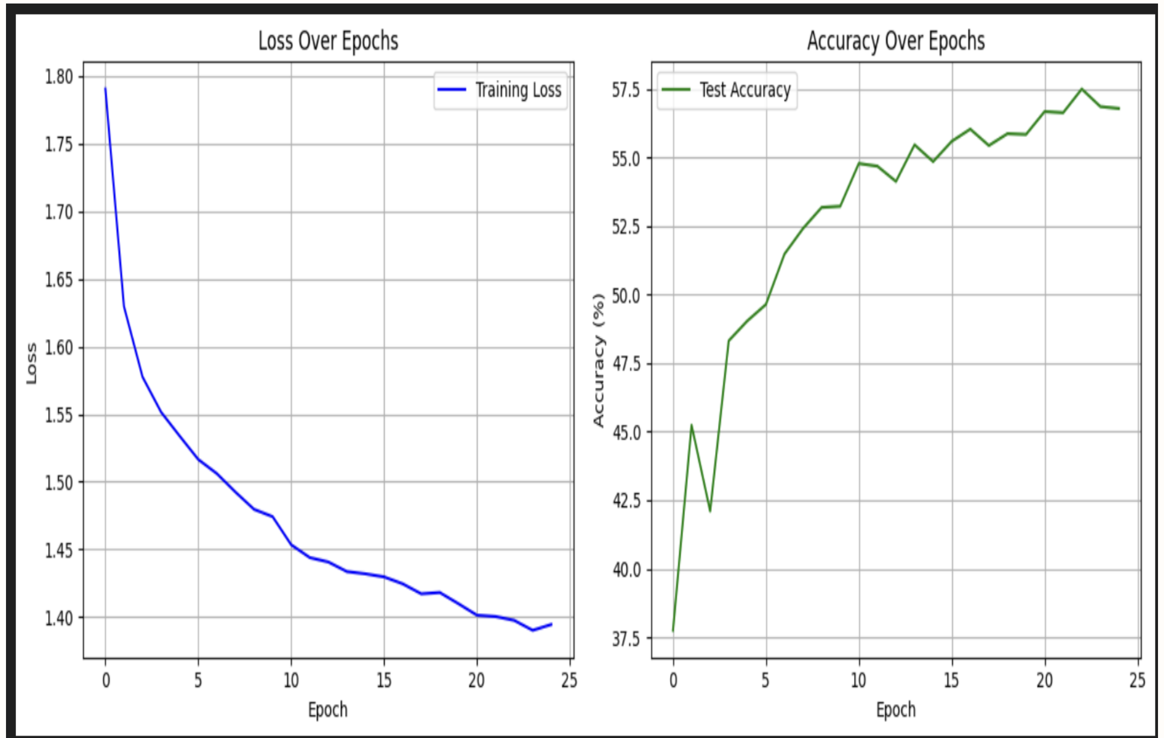
Sustained 58.57% accuracy with smoother training curves.

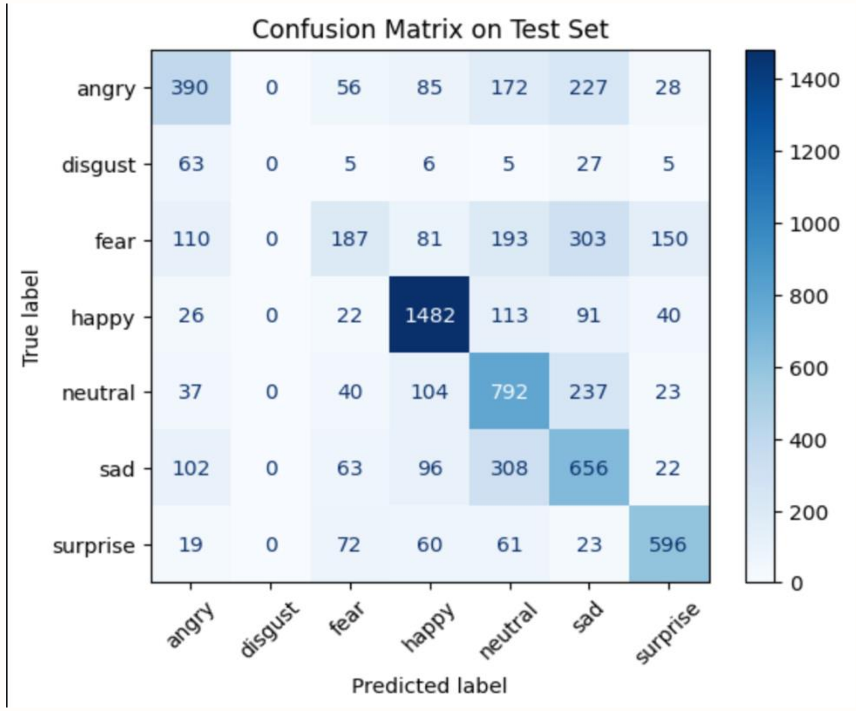
Class-wise Improvements

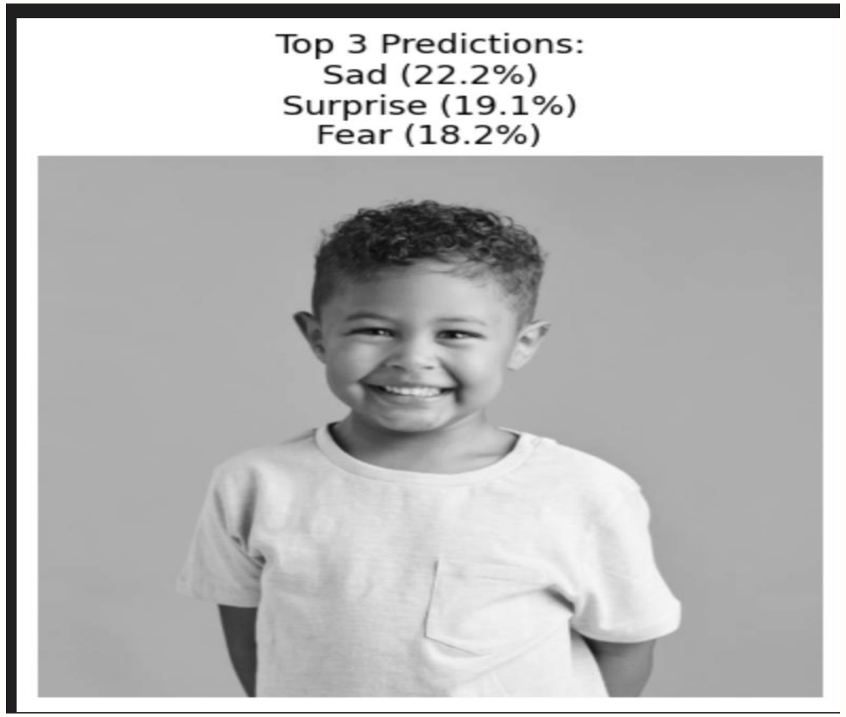
More balanced recognition across all emotion classes.

CLAHE Impact

Enhances local facial features critical for emotion detection.







Results - ResNet18 Pretrained

Test Accuracy

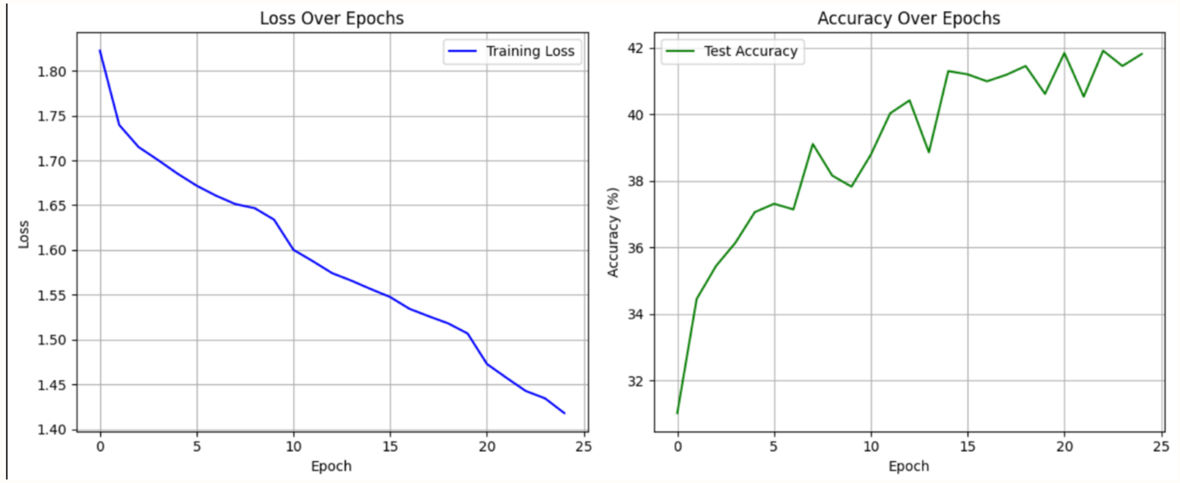
Underperformed with only 41.90% accuracy.

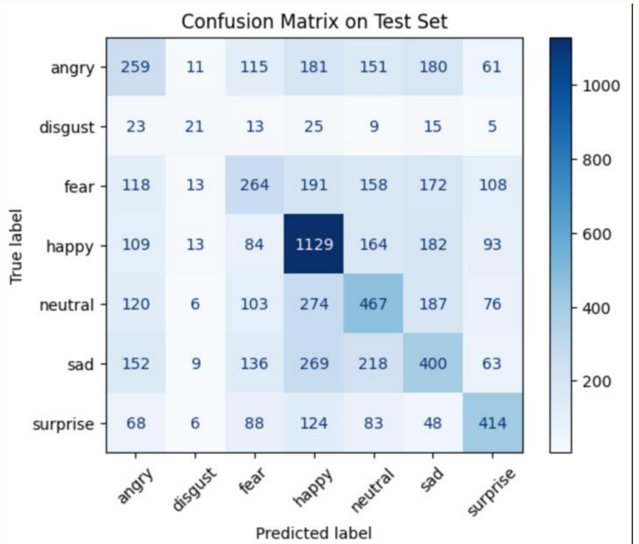
Good Class

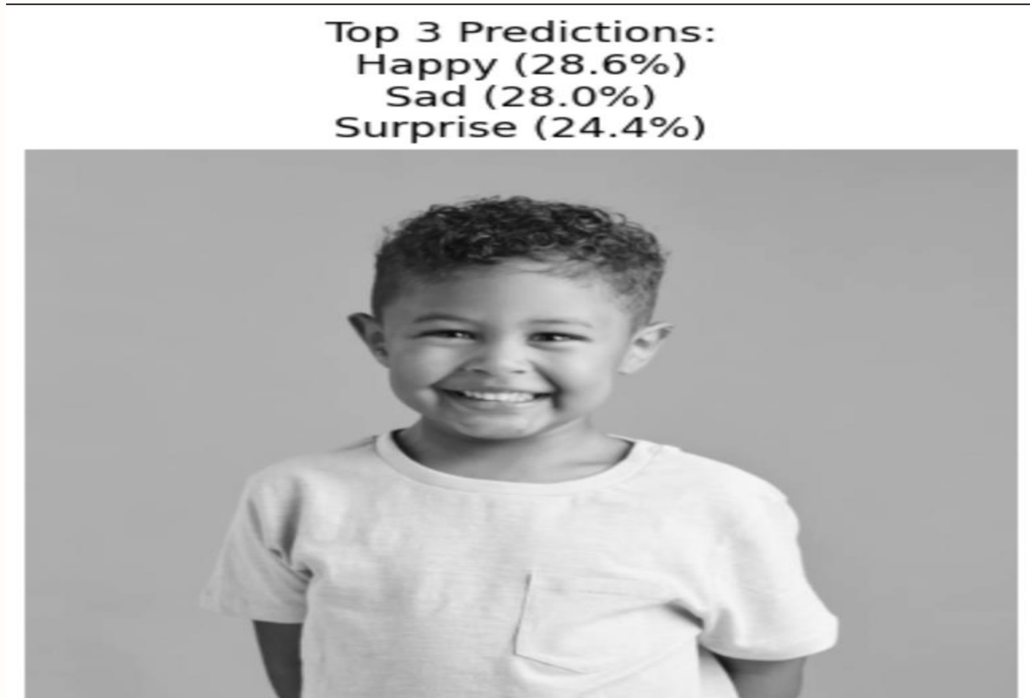
Strongest on 'happy' emotion recognition.

Limitations

- Domain mismatch from RGB to grayscale
- Poor generalization on FER2013 data







Future Work

1 Expand Datasets

To further validate the performance of our models, we plan to extend our experiments to larger and more diverse emotion recognition datasets such as FER+ and AffectNet.

2 Incorporate Facial Landmarks

Incorporating facial landmark information could provide valuable spatial cues to improve the models' understanding of facial expressions.

3 Improve Interpretability

Integrating techniques like Grad-CAM or LIME would allow us to better interpret the inner workings of our models and gain insights into the features they are learning.

4 Explore Transformer Models

Given the recent success of transformer-based architectures in computer vision, we aim to investigate their potential for emotion recognition tasks. Additionally, we will explore real-time deployment of our models on edge devices.

Thank You!